

Photoinitiated Thiol–Ene Click Reaction for Preparation of Highly Adhesive and Mechanically Stable Silicone Coatings for Marine Antifouling and Anticorrosion

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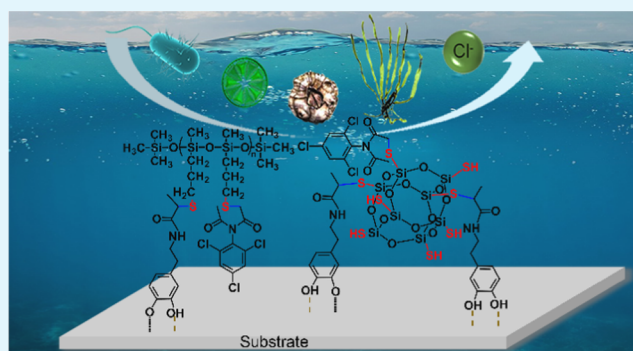
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ABSTRACT: Marine biofouling and corrosion have become the main problems affecting the development of the marine industry. Silicone-based coatings have been widely used for antifouling and anticorrosion due to their low surface energy. However, the poor adhesion and low mechanical stability of these materials limit their application in complex marine environments. In this work, we presented a marine antifouling and anticorrosion coating named POSS-DMA@PDMS-TCM through photoinitiated thiol–ene click reaction combined with (mercaptopropyl) methylsiloxane dimethylsiloxane (PDMS-SH), dopamine methacrylamide (DMA), sulfhydryl-functionalized organosiloxanes (POSS-SH)₈, and *N*-(2,4,6-trichlorophenyl) maleimide (TCM). The POSS-DMA@PDMS-TCM coating exhibited strong stability and bonding ability both in air (2.17 MPa) and underwater (2.11 MPa) when the DMA content was 3 wt %. The high antibacterial (98.1% for *Staphylococcus aureus* and 99.5% for *Escherichia coli*) and antidiatom (94.5%) properties of the POSS-DMA@PDMS-TCM coatings have also been confirmed. Moreover, the POSS-DMA@PDMS-TCM coatings show excellent antifouling abilities in 120-day marine field tests, reducing fouling by 65.5% in comparison to the blank group. The coating also displayed superior anticorrosion performance with E_{corr} values of -0.055 V, I_{corr} values of 7.67×10^{-6} , and R_p values of $3.10 \times 10^5 \Omega$ for Cu, which benefited from excellent chelating effect and liquid repellency. This study provides a novel strategy for the development of high-quality marine antifouling and anticorrosion coatings.

KEYWORDS: PDMS, click reaction, adhesion strength, antibiofouling, anticorrosion



1. INTRODUCTION

Marine biofouling, caused by the accumulation of marine plants and algae and colonization of marine organisms, is a global issue affecting marine and aquaculture industries.^{1,2} Fouling organisms that adhere to submerged surfaces can increase hydrodynamic drag, impair ship maneuverability, boost fuel consumption, block circulation pipes, and compromise instrumentation, leading to significant economic maintenance costs and environmental safety concerns.^{3–5} In addition, microorganisms or macroorganisms on metal surfaces accelerate the corrosion reaction by changing the electrochemical conditions at the interface.⁶ The detrimental influence of microbiologically induced corrosion can intensify metal degradation, leading to additional economic losses.^{7–9} Antifouling coatings are widely seen as a potent solution to this problem. Conventional coatings often contain toxic substances, such as arsenic, sulfur (S), mercury, and tributyltin (TBT).^{10,11} Consequently, efforts to develop eco-friendly yet effective alternatives are growing, such as silicone-based fouling-release coatings, amphiphilic copolymers, protein-resistant polymers, degradable polymers, and biomimetic polymers.^{2,12–15}

Among these materials, poly(dimethylsiloxane) (PDMS) is continuously gaining attention as an antifouling coating due to its low surface energy, low Young's modulus, and relatively economical cost.¹⁶ Silicone-based antifouling coatings generally leverage the hydrophobicity, low surface energy, and elastic modulus of the PDMS matrix.^{13,17–19} Furthermore, PDMS coatings are an excellent corrosion protection material and extend the service life of steel structures.²⁰ Fouling organisms adhered to submerged surfaces can be easily dislodged by using hydraulic shear force. However, the mechanical stability and adhesion of PDMS are poor in practical applications due to its low modulus, nonpolarity, and inertness.²¹ Consequently, it is essential to enhance mechanical

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stability and adhesion. The modification of a linear PDMS elastomer through the employment of diverse groups, namely, epoxy, catechol, urethane, urea, and 2-ureido-4[1H]-pyrimidine, has the potential to augment the substrate adhesion as well as the mechanical properties within the context of silicone coatings.^{22–25} However, such modifications often compromise their antifouling ability due to surface alterations caused by the introduction of polar groups.²⁶ The use of inorganic particles (such as SiO₂, nanodiamonds, graphene oxide, and multiwalled carbon nanotubes) to improve substrate adhesion frequently increases PDMS coating surface roughness, thus weakening its antifouling performance.^{27–29} Thus, achieving an equilibrium among outstanding antifouling capabilities, anticorrosion characteristics, substrate adhesion, and mechanical strength poses a substantial challenge in the realm of silicone coatings.

Recently, extensive investigations have been carried out regarding the 3,4-dihydroxyphenylalanine (DOPA) existing in mussel adhesion proteins, which is attributed to its remarkable adhesive characteristics.³⁰ A multitude of antifouling coatings have been devised through the utilization of catechol groups as biomimetic anchoring moieties.^{31,32} These coatings demonstrate long-term stability due to the presence of catechol groups and the expected antifouling performance. While the application of catechol-containing amphiphilic coatings on antifouling surfaces has been examined, the use of catechol derivatives within PDMS components to improve adhesive performance through polymerization has not been investigated. Several studies have shown that cross-linked polymer structures with catechol functionality can be successfully fabricated through photopolymerization reactions.³³ Furthermore, polyhedral oligomeric silsesquioxane (POSS) is a class of cage-shaped inorganic–organic hybrid chemical molecules that has developed rapidly in recent years.^{34–36} The introduction of POSS into polymer matrices enhances their mechanical properties. However, to the best of our knowledge, the use of POSS to enhance PDMS-based marine antifouling coatings has rarely been reported.

In this study, an excellent PDMS-based antifouling and anticorrosion coating with high adhesion and excellent mechanical properties was prepared by the thiol–ene click reaction. Specifically, dopamine methacrylamide (DMA) and polyhedral oligomeric silsesquioxane (POSS) were used to improve the adhesion and mechanical strength of the PDMS coating; N-(2,4,6-trichlorophenyl) maleimide (TCM) was chosen as an environmentally friendly antifouling agent with low toxicity. In addition, the thiol–ene click reaction was introduced to facilitate the preparation of the coating. The adhesion strength and surface wettability of the coatings were subject to the influence of the DMA content therein. Different POSS contents were tested for their influence on the mechanical stability of the surface. The resulting coating, named POSS-DMA@PDMS-TCM, was further characterized, and its performance in adhesion, mechanical strength, stability, antifouling, and anticorrosion was evaluated.

2. EXPERIMENTAL SECTION

2.1. Materials. PDMS-SH number-average molecular weight = 5500 g·mol^{−1}, polydispersity index = 1.40, mol % (mercaptopropyl) methylsiloxane = 100, was purchased from Gelest Inc. (North Carolina) and used without any purification. Sodium carbonate (analytical reagent [AR]), sodium tetraborate (99%), dopamine hydrochloride (98%), tetrahydrofuran (THF, AR), sodium hydroxide (NaOH, AR), ethyl acetate (EAC, 99.5%), magnesium sulfate

(MgSO₄, AR), and hexane (C₆H₁₄, AR) were purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). Then, 2,2-dimethoxy-2-phenylacetophenone (99%) was obtained from Sigma-Aldrich. Finally, (3-mercaptopropyl) trimethoxysilane (97%), hydrochloride (HCl, 37%), methanol (MeOH, AR), and toluene (≥99.5%) were purchased from Macklin Reagent Co., Ltd. (Shanghai, China). Other analytical-grade reagents were obtained and used as received.

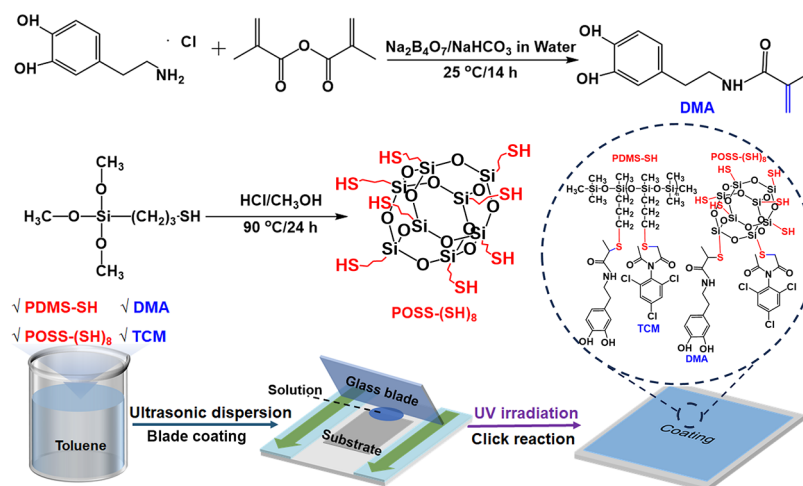
2.2. Synthesis of DMA. The DMA was synthesized according to the methodology in the literature.³⁷ First, sodium borate (10 g) and sodium bicarbonate (4 g) were dissolved in deionized water (100 mL) under magnetic stirring. Subsequently, nitrogen was bubbled through the solution for 20 min. 3,4-dihydroxyphenethylamine hydrochloride (5 g) was introduced into the reaction system and thoroughly dispersed. Subsequently, methacrylate anhydride (4.7 mL, 58.1 mmol) was added. To maintain the pH above 8, a 1 mol/L NaOH solution was employed for pH adjustment. Thereafter, the reaction mixture was stirred overnight at room temperature under a nitrogen atmosphere. The mixture underwent two washing cycles with ethyl acetate (50 mL). The pH of the mixture was meticulously adjusted to below 2 through the addition of a 6 M HCl solution. The mixture was extracted three more times with ethyl acetate (50 mL). The volume of the organic layer was reduced to 25 mL and dried with MgSO₄. The brown solid was precipitated with vigorous stirring in hexane. The solid was further purified using ethyl acetate and hexane and then dried under a vacuum. The suspension was kept at 4 °C overnight. ¹H NMR (600 MHz, DMSO-*d*₆, δ): 8.90 (OH, H), 8.29 (NH, 1H), 6.43 (Ph, 1H), 5.51 (HC=C, 1H), 3.31 (NH-CH₂-, 2H), 2.58 (CH₂-Ph, 2H), 1.91 (CH₃, 3H).

2.3. Synthesis of POSS-(SH)₈. The synthesis of POSS-(SH)₈ was based on a previous study.³⁸ First, (3-mercaptopropyl) trimethoxysilane (30 mL) and HCl (60 mL) were combined with MeOH (500 mL) and then stirred at 90 °C. A white precipitate was generated after 24 h. Finally, POSS-(SH)₈ was obtained after separation and purification. ¹H NMR (600 MHz, DMSO-*d*₆, δ): 2.38 (SH-CH₂, 2H), 2.13 (CH₂, 2H), 1.50 (SH, 1H), 0.61 (Si-CH₂, 2H).

2.4. Fabrication of PDMS-Based Coating. PDMS-SH, POSS-(SH)₈, DMA, TCM, and DMPA were mixed with 2 mL of toluene. This mixture was completely dissolved via ultrasonic treatment and became a homogeneous solution. The mixture solution was applied to the substrate via the blade coating method. After the coating process, the samples were placed under UV light (365 nm, 1.3 W·cm^{−2}) for 5 min. The separation distance between the coating and the UV light source was approximately 10 cm. ¹H NMR (600 MHz, DMSO-*d*₆, δ): 9.46 (OH, 1H), 8.07 (NH, 1H), 7.68 (Ph, 1H), 6.41 (Ph, 1H), 3.74 (–S–CH, 1H), 3.43 (NH–CH₂–, 2H), 2.88 (CNO–CH₂–, 2H), 2.49 (–S–CH₂–, 2H), 2.62 (CH₂–Ph, 2H), 1.89 (SH, 1H), 1.48 (CH₂, 2H), 0.77 (Si–CH₂–, 2H), 0.68 (CH₃, 3H), 0.34 (Si–CH₃, 3H), 0.20 (Si–CH₃, 3H).

2.5. Characterization. The surface morphology and cross-sectional thickness of the coatings were examined through the utilization of field-emission scanning electron microscopy (FESEM; Regulus-8100, Hitachi). The surface roughness of each coating was characterized by using a surface roughness tester (SJ-210, Mitutoyo) according to ISO 1997. The success of the reaction was confirmed by Fourier transform infrared spectroscopy (FTIR; Nicolet 6700, Thermo Fisher Scientific). Surface elemental analysis was performed by utilizing X-ray photoelectron spectroscopy (XPS; K-Alpha+, Thermo Fisher Scientific) with Al Kα radiation, and the binding energy was calibrated using C 1s (284.80 eV). The ¹H NMR spectra were acquired with the employment of an AVANCE 600 MHz spectrometer (Bruker), wherein tetramethylsilane served as the internal standard. Meanwhile, POSS-(SH)₈, DMA, and POSS-DMA@PDMS-TCM coatings were dissolved in DMSO-*d* at ambient temperature to prepare samples suitable for NMR analysis. The contact angle (CA) of deionized water on each coating was quantified using a CA meter (Theta Lite, Biolin Scientific). The average value was computed based on five randomly selected area measurements for each coating. The surface free energy (SFE) was determined through

Scheme 1. Schematic Diagrams of Synthesis Routes of DMA, POSS-(SH)₈, and POSS-DMA@PDMS-TCM Antifouling Coating



the application of the Owens–Wendt–Rabel–Kaelble method with the utilization of the following equation.³⁹

$$\gamma_s = \gamma_s^d + \gamma_s^p \quad (1)$$

$$\gamma_L(1 + \cos \theta) = 2(\gamma_s^d \gamma_L^d)^{1/2} + 2(\gamma_s^p \gamma_L^p)^{1/2} \quad (2)$$

For water, $\gamma_L^d = 21.8$ mN/m and $\gamma_L^p = 51.0$ mN/m. For diiodomethane, $\gamma_L^d = 48.5$ mN/m and $\gamma_L^p = 2.3$ mN/m. γ_s^d represents the dispersion component of the SFE. γ_s^p represents the polar component of the SFE.

2.6. Antibiofouling Test. Each sample was submerged in 10 mL of a *Staphylococcus aureus* or *Escherichia coli* suspension ($\sim 10^6$ colony-forming units) and incubated at 37 °C for 12 h. Subsequently, the samples underwent three successive rinsing cycles with PBS to remove unstable adherent bacteria. The samples were subjected to multiple rinsing procedures to achieve a complete separation of the bacteria that had attached to the surfaces. Subsequently, 200 μ L of the PBS rinse containing the dislodged bacteria was evenly dispersed across the surface of the Luria–Bertani agar plates. These plates were then placed in an incubator maintained at a constant temperature of 37 °C for 18 h.

The samples were submerged within a suspension of *Nitzschia closterium f. minutissima* and then subjected to incubation for 7 days at a constant temperature of 20 °C, under a 12:12 h light–dark cycle regime. The diatoms that were unstably attached to the surface were eliminated by using rinsing with seawater. For the extraction of diatom chlorophyll, the samples were submerged in 80 wt % aqueous acetone solution. Subsequently, the fluorescence intensity was quantified by employing an Amiscience fluorometer. The aforementioned experimental procedures were replicated on three separate occasions to ensure the reproducibility and reliability of the obtained results.

The antibacterial and antidiatom rate (R) was calculated as

$$R = \frac{N_b - N_c}{N_b} \times 100\% \quad (3)$$

where N_b and N_c represent the quantities of bacterial colonies and diatoms, respectively.

The bacterial solution was transferred from the coatings onto the silicon wafers. Subsequently, the silicon wafers were immersed in a 2.5% glutaraldehyde solution for 1 min. The silicon wafers were washed twice through gentle immersion in a PBS solution. Subsequently, the wafers were successively submerged in four standard ethanol–water solutions, namely, 10, 50, 90%, and concentrated ethanol, each for 1 min. Scanning electron microscopy (SEM) analysis was performed on air-dried wafers to examine the morphology of bacterial cells.

2.7. Marine Field Tests. A 120-day marine field test was carried out from September to December (2023) in the Bohai Sea (117°46'E, 39°43'N) in Tianjin, China. Epoxy and stainless steel plate specimens (50 mm $\times$ 50 mm $\times$ 3 mm) were submerged in seawater at a depth of 0.5 m. Periodically, the surfaces of the samples were cleansed by a gentle rinse with seawater to eliminate unstable adhering dirt. Subsequently, the samples were photographed by using a camera to document the surface conditions. Quantitative analyses were performed using ImageJ software.

2.8. Electrochemical Measurements. Electrochemical measurements (Tafel polarization analysis and electrochemical impedance spectroscopy (EIS)) were executed employing a standard three-electrode system (CHI660C, CH Instruments). The coated metal, platinum foil, and the Ag/AgCl electrode (1 M, KCl) were selected as the working electrode, the counter electrode, and the reference electrode, respectively. The electrolyte was a 3.5 wt % NaCl solution. Tafel curve tests were performed with a scan rate of 0.01 V·s⁻¹, and the corrosion current density (I_{corr} , A·cm⁻²), polarization resistance (R_p , Ω ·cm²), and corrosion potential (E_{corr} , V) were analyzed using CHI 660e software. The EIS was measured at 10⁻⁵–10⁻² Hz with a perturbation voltage of 20 mV. The EIS plots were fitted using ZSimpWin software.

3. RESULTS AND DISCUSSION

3.1. Structural and Morphological Characterization.

First, DMA and POSS-(SH)₈ were prepared as strengthening agents for the coatings. As shown in Scheme 1, DMA was prepared using a modified version of a previously reported technique under moderately basic conditions.⁴⁰ DMA was demonstrated to augment the adhesion properties of the coating. POSS-(SH)₈ can improve the mechanical strength of the coating due to its caged siloxane structure. The thiol–ene click reaction served as an effective means to firmly integrate the constituent components. It was preliminarily concluded that the PDMS-based coating could be prepared on various substrate materials (Figure S1). TCM was selected as an environmentally friendly organic marine antifouling agent with low toxicity, giving coatings excellent antifouling properties.⁴¹

The surface morphologies of the DMA@PDMS, POSS-DMA@PDMS, and POSS-DMA@PDMS-TCM coatings were observed via scanning electron microscopy (SEM). The DMA@PDMS coating displayed a relatively smooth topography (Figure S2a) due to PDMS cross-linking. Raised dots of different sizes were randomly distributed on the coating surface because of the embedded POSS (Figure S2b). The

number and volume of raised spots on the coating surface further increased with the addition of TCM (Figure 1a). The

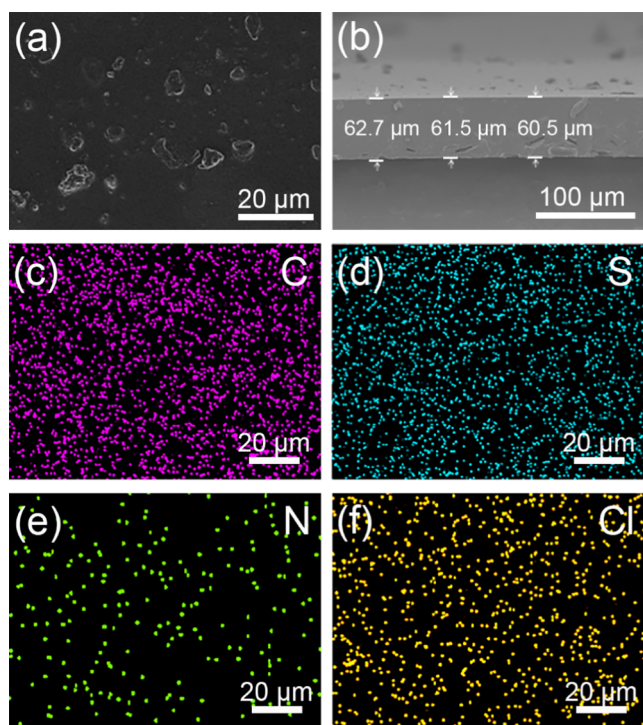


Figure 1. (a) Morphology characterization of POSS-DMA@PDMS-TCM coating. (b) Cross-sectional SEM image of POSS-DMA@PDMS-TCM coating. (c–f) C, S, N, and Cl element distribution maps of POSS-DMA@PDMS-TCM coating.

SEM-measured thickness of the POSS-DMA@PDMS-TCM coating was approximately 61.5 μm (Figure 1b). Energy-dispersive X-ray analysis showed that elements C, S, N, and chlorine (Cl) were well distributed on the POSS-DMA@PDMS-TCM coating surface (Figure 1c–f).

POSS-(SH)₈ and DMA were synthesized according to a method described in the literature. The synthesized POSS-(SH)₈ and DMA monomers were characterized via FTIR. As shown in Figure 2a, FTIR analysis of POSS-(SH)₈ revealed an absorption peak positioned at 1112 cm^{-1} , which corresponded to the stretching vibration of the Si–O–Si skeleton in the caged sesquicarbazon, indicating the formation of the POSS cage. The absorption peaks at 690, 1259, and 2557 cm^{-1} corresponded to the out-of-plane bending vibrations of Si-(CH₃)₂, the symmetric deformation vibrational absorption of Si-(CH₃)₂, and the stretching vibration of –SH, respectively. The absorption band at 3189 cm^{-1} was attributed to the –OH stretching vibration, whereas the peak at 1594 cm^{-1} corresponded to C=C stretching. The absorption band at 3397 cm^{-1} was attributed to –NH stretching vibration. The absorption bands at 1396 and 1191 cm^{-1} were assigned to the benzene ring. Thus, FTIR demonstrated the success of the POSS-(SH)₈ and DMA monomer preparation. The peaks at 2557 and 1594 cm^{-1} were assigned to the thiol groups and typical C double bond in PDMS-SH, POSS-(SH)₈, DMA, and TCM, respectively, which disappeared after photo-cross-linking in the POSS-DMA@PDMS-TCM coating.

XPS analysis was also performed to study the reaction process between POSS-(SH)₈, PDMS-SH, DMA, and TCM. The elements C, N, and O were detected on the DMA surface, whereas Si, S, and Cl peaks were detected on the POSS-DMA@PDMS-TCM surface (Figure 2b), demonstrating the successful formation of the POSS-DMA@PDMS-TCM coating. Furthermore, Figure 2c shows the C peak for DMA,

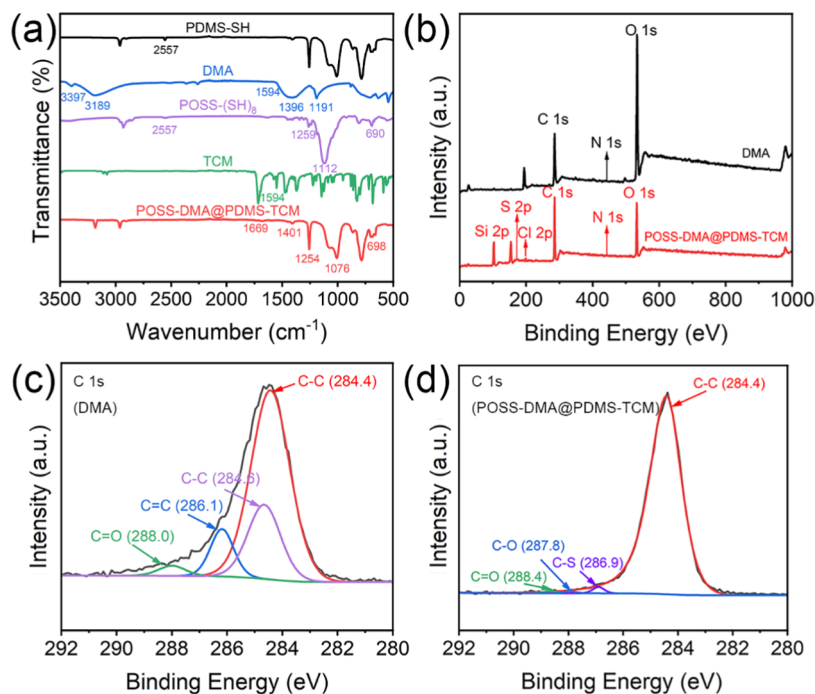


Figure 2. (a) FTIR spectra of PDMS-SH, DMA, POSS-(SH)₈, TCM, and POSS-DMA@PDMS-TCM coating. (b) XPS survey scan spectra of prepared DMA and POSS-DMA@PDMS-TCM coating. High-resolution XPS spectra of (c) C 1s of DMA and (d) C 1s of POSS-DMA@PDMS-TCM coating.

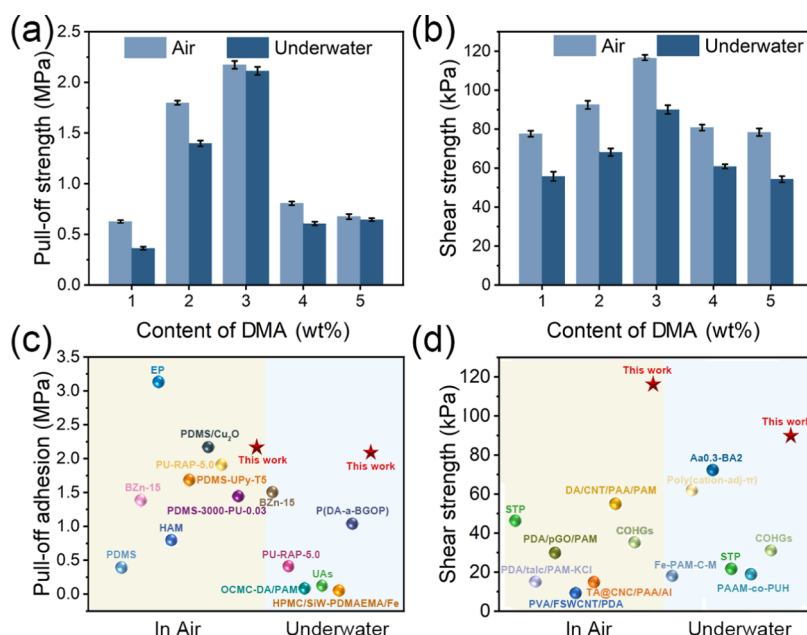


Figure 3. Strong substrate bonding ability of the POSS-DMA@PDMS-TCM coating. (a) Pull-off adhesion and (b) shear strength of POSS-DMA@PDMS-TCM coatings on steel substrates in air and underwater. (c) Pull-off adhesion and (d) shear strength of the POSS-DMA@PDMS-TCM coating and previously reported adhesives and coatings.

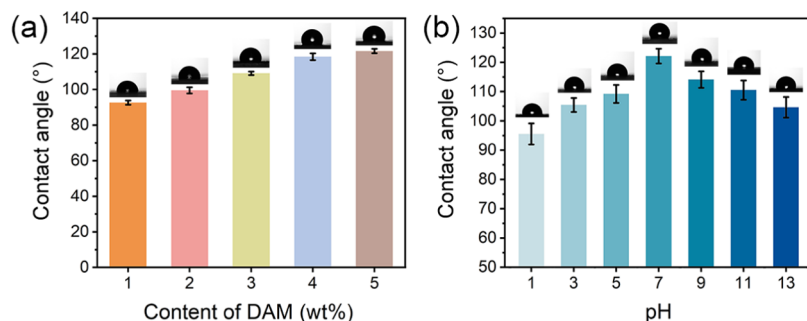


Figure 4. Wetting properties of POSS-DMA@PDMS and POSS-DMA@PDMS-TCM coatings. (a) Water CA profiles on POSS-DMA@PDMS coatings with different DMA contents. (b) CA distribution of water droplets on POSS-DMA@PDMS-TCM coatings at pH values of 1, 3, 5, 7, 9, 11, and 13.

resolved into four components centered at 288.0, 286.1, 284.6, and 284.4 eV, which were ascribed to the C=O, C–O, C–C, and C–H species, respectively. A new peak centered at 286.9 eV was detected on the POSS-DMA@PDMS-TCM surface (Figure 2d), which was attributed to C–S bonding, indicating the grafting of the POSS-DMA@PDMS-TCM coating.

3.2. Adhesive Properties. The adhesion characteristics of the coating concerning the substrate constitute a fundamental and indispensable prerequisite for the overall performance of the coating. In this study, DMA was employed as an adhesive agent to significantly enhance the adhesion performance of the coating. The bond strength of the coatings was tested in air and underwater, respectively. The experimental results are presented in Figure 3a. Remarkably, the POSS-DMA@PDMS-TCM coatings exhibited strong bonding to their substrates in the air and underwater. The pull-off adhesion of the POSS-DMA@PDMS-TCM coatings increased and then decreased with an increase in the DMA content, both in air and underwater. At a DMA content of 3%, the pull-off adhesion of the POSS-DMA@PDMS-TCM coating was the highest, at 2.17 MPa in air and 2.11 MPa underwater. At a DMA content of 2%, the adhesion of the coating decreased but remained

superior to that of the POSS-DMA@PDMS-TCM coatings with other DMA contents. Shear adhesion followed the same pattern as the variation in pull-off adhesion (Figure 3b). In addition, the effect of the TCM content on the coating adhesion strength was explored. Figure S4 shows the pull-off strength and shear strength of POSS-DMA@PDMS coatings with different TCM contents on steel substrates. The pull-off strength and shear strength of POSS-DMA@PDMS were approximately 2.32 MPa and 125.5 kPa in air, respectively. The pull-off strength and shear strength of POSS-DMA@PDMS underwater were approximately 2.25 MPa and 100.5 kPa, respectively. As the antifoulant content increased, the adhesion gradually decreased. Anyhow, the POSS-DMA@PDMS-TCM samples demonstrated an adhesion strength exceeding 1.0 MPa, which fulfills the requisite criteria for practical applications within the marine environment.⁴² The strong bonding capability of the POSS-DMA@PDMS-TCM coatings could be attributed to the introduction of the DMA adhesive and the high cross-linking degree between DMA, PDMS-SH, and POSS-(SH)₈ (Scheme 1). The adhesion strength of the POSS-DMA@PDMS-TCM coating in air and underwater was better than that of the other reported

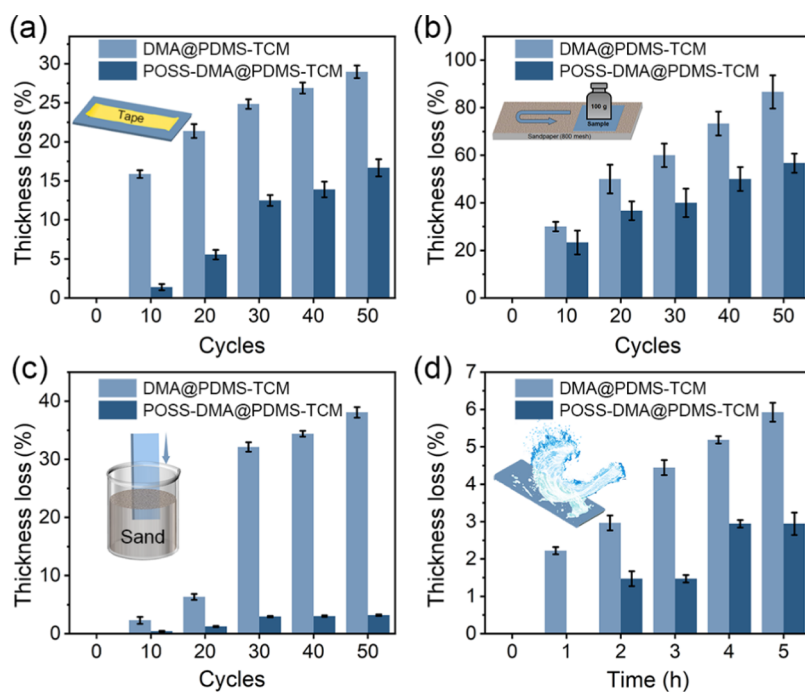


Figure 5. Mechanical durability analysis. Thickness loss rate after (a) tape peeling, (b) sandpaper abrasion cycles, (c) sand abrasion cycles, and (d) water jetting on two different coated surfaces.

coatings and adhesives (Figure 3c,d and Tables S1 and S2).^{23,24,34,43–64} Interestingly, the POSS-DMA@PDMS-TCM coating had strong bonding with a glass substrate capable of lifting weights of up to 5 kg (Figure S5). Therefore, the POSS-DMA@PDMS-TCM coating had exceptional bond strength with substrates.

3.3. Surface Wettability. The resulting PDMS-based coating displayed water-repelling properties in air. As shown in Figure 4a, the CA of the POSS-DMA@PDMS coatings increased with the DMA content, all of which were greater than 90°. This phenomenon was attributed to the low surface energy of PDMS, which made the coatings hydrophobic. Additionally, coating roughness increased with the DMA content, thus enhancing the CA and hydrophobicity of the coatings (Figure S6). The incorporation of TCM led to an augmentation in the water contact angle of the coating. This enhancement can be attributed to the hydrophobic nature of the phenyl group present within the TCM molecule.⁶⁵ TCM further improved the hydrophobicity of the coatings, thereby enhancing their antifouling and anticorrosion properties, which will be discussed later. The hydrophobicity of the coatings, combined with the excellent chemical stability of PDMS, as shown in Figure 4b, rendered the POSS-DMA@PDMS-TCM coatings resistant to liquids with pH values ranging from 1 to 13, with their CA consistently exceeding 90°. Even after full submersion in dyed water for 1 day, their hydrophobicity remained stable, allowing the PDMS-based coating surface to be completely cleaned without any water penetration, as illustrated in Figure S7. These results demonstrated the strong hydrophobicity of the POSS-DMA@PDMS-TCM coating, which enhanced the coating's long-term impermeability and antifouling ability in seawater.

3.4. Mechanical Stability. For practical applications, the robustness of antifouling coatings warrants a comprehensive and in-depth investigation. The cage-like polysilsesquioxane (POSS), owing to the advantageous Si–O–Si inorganic core

structure it possesses, enhances the mechanical strength of the coating.^{66,67} In the current study, a concentration of 2 wt % of POSS-(SH)₈ was selected for conferring mechanical reinforcement to the coating. Tests were performed to systematically evaluate the coating robustness to tape peeling, sandpaper abrasion cycles, sand abrasion cycles, and water jetting, as shown in Figure 5. As shown in Figure 5a, one peel-off was considered to be one cycle of tape adhesion. In the tape adhesion experiment, each coated surface was evaluated using thickness loss measurements after every 10 cycles, and this process was repeated five times. After the tape adhesion experiment, the thickness loss of the POSS-DMA@PDMS-TCM coating was significantly lower than that of the DMA@PDMS-TCM coating. Linear sandpaper abrasion tests were also performed, as shown in Figure 5b. Each coated substrate was placed under a (P800) sandpaper and a 100 g load; then, the sandpaper was moved on the coated surface for 10 cm back and forth, which constituted one abrasion cycle. The experimental results showed that after 50 abrasion cycles, the thickness loss rate of the POSS-DMA@PDMS-TCM coating was reduced by nearly 40% relative to that of the POSS-DMA@PDMS coating. We used a more rigorous approach to further investigate the mechanical stability of the coatings. Glass substrates coated with the PDMS-based coatings were inserted into a sand pile (90 mesh) and then lifted, constituting one sand abrasion cycle (Figure 5c). The thickness loss of the POSS-DMA@PDMS coating was as high as 38%, whereas that of the POSS-DMA@PDMS-TCM coating was only 3.1%. Furthermore, as shown in Figure 5d, the POSS-DMA@PDMS-TCM coating retained its excellent mechanical stability after the water jetting test (0.071 MPa water pressure), as verified by the change in the thickness loss rate. As shown in Figure S8, the mechanical stability of the DMA@PDMS-TCM and POSS-DMA@PDMS-TCM coatings was further investigated by analyzing the CA change during tape peeling, sandpaper abrasion, sand abrasion, and water jetting tests. The CAs of the

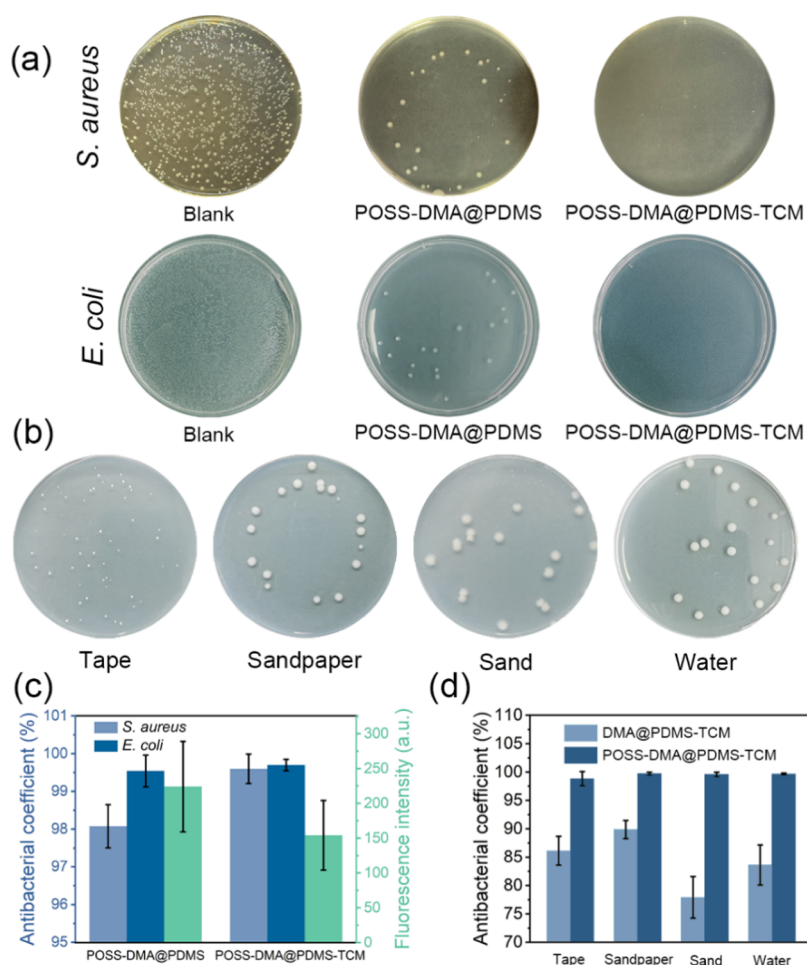


Figure 6. Assessment of antibiofouling performance. (a) Colonial morphology images of *S. aureus* and *E. coli* attached to POSS-DMA@PDMS and POSS-DMA@PDMS-TCM coatings. (b) Colonial morphology images of POSS-DMA@PDMS-TCM coating after tape stripping, sandpaper abrasion cycle, sand abrasion cycle, and water jetting. (c) Antibacterial coefficient and fluorescence intensity of POSS-DMA@PDMS and POSS-DMA@PDMS-TCM coatings. (d) Antibacterial (*E. coli*) coefficients of DMA@PDMS-TCM and POSS-DMA@PDMS-TCM coatings after mechanical damage.

DMA@PDMS-TCM and POSS-DMA@PDMS-TCM coatings gradually decreased with an increase in the experimental period. The decreasing trend of the CA of the DMA@PDMS-TCM coating was more evident than that of the POSS-DMA@PDMS-TCM coating. This was consistent with the thickness loss rate measurements. The great mechanical strength of the POSS-DMA@PDMS-TCM coating highlighted its potential practical antifouling applications in ships and marine working devices.

3.5. Antibiofouling Properties. Bacterial attachment is a key step in promoting the accumulation and attachment of diatoms, algae, and barnacles. It is also the first step in regulating their irreversible attachment after membrane formation. Antimicrobial properties are an aspect of antifouling that can terminate the fouling process at an early stage and can be used to assess the antifouling performance of a coating. *S. aureus* (Gram-positive bacteria) and *E. coli* (Gram-negative bacteria) were selected as representatives of the broad-spectrum antibacterial evaluation. As shown in Figure 6a, the POSS-DMA@PDMS coating significantly inhibited the survival of *S. aureus* and *E. coli*. The antibacterial coefficient rates of the POSS-DMA@PDMS coating against *S. aureus* and *E. coli* were 98.1 and 99.5%, respectively (Figure 6c). As shown in Figure S14, the POSS-DMA@PDMS coating had a surface

free energy of 26.17 mN/m. PDMS segments with low surface energy facilitated biofouling removal. The POSS-DMA@PDMS-TCM coating, which had added TCM, maintained a high antimicrobial efficiency. The antimicrobial coefficients of the POSS-DMA@PDMS-TCM coating were 99.7 and 99.6% against *S. aureus* and *E. coli*, respectively. The number of bacteria attached to the surface of the coating decreases as the antifouling agent content increases. SEM images revealed visible ruptures in the cell membranes of *E. coli* on the POSS-DMA@PDMS-TCM coating surfaces (Figure S9), which could further suggest the biocidal action of the TCM. *N. closterium f. minutissima* was chosen as the model to further evaluate the antibiofouling performance of the coatings. The fluorescence intensity of the diatoms attached to the coatings, which were immersed in an artificial seawater solution for 7 days, was determined. As shown in Figure 6c, the fluorescence intensity of the diatoms attached to the POSS-DMA@PDMS-coated surface (224 AU) was significantly lower than that of the control (2776 AU) after 7 days of immersion in the *N. closterium f. minutissima* solution. After the addition of the TCM, the POSS-DMA@PDMS-TCM coating (154 AU) showed outstanding antidiatom performance compared to the POSS-DMA@PDMS coating. In brief, the TCM groups acted as contact killers.

Bacteria resistance and sterilization capabilities are essential for inhibiting bacterial attachment to and proliferation on surfaces. The antifouling performance of the coating surfaces was more intuitively observed after the mechanical stability test. Their antimicrobial performance was assessed after 50 tape strippings, 50 sandpaper abrasion cycles, 50 grit abrasion cycles, and 5 h of water-flow impact. As shown in Figure 6b, the antimicrobial resistance (*E. coli*) of the DMA@PDMS-TCM coating decreased to above 90% after the series of mechanical stability tests described above. The antimicrobial rate of the coating after tape stripping, sandpaper abrasion, grit abrasion, and grit abrasion decreased to 86.13, 89.86, 77.91, and 77.91%, respectively, which indicated that the surface of the coating was severely damaged. However, the antimicrobial rates of the POSS-DMA@PDMS-TCM coating remained above 95% after the same mechanical stability test. This is attributed to the nanostructure of POSS, which improves the mechanical stability of the coating and thus maintains its excellent antimicrobial performance.

3.6. Marine Field Antifouling. The process of marine biofouling is complex due to the wide variety of organisms and different time spans.⁶⁸ Marine field tests were conducted in the Bohai Sea, China, to verify the antifouling properties and stability of the coatings. Typical images of the tested panels after immersion in seawater for 4 months are shown in Figures 7a and S10. ImageJ was used to process the photos in Figures S11 and S12. The corresponding fouling coverage areas are shown in Figure 7b. After 1 month, the control plates exhibited adhesion of tubeworm larvae and diatoms, among others. After 2 months, more severe biofouling was observed on the control panels, including attachment of barnacles and mussels. In contrast, there was almost no biofouling on the surface of the

POSS-DMA@PDMS-TCM coating. The POSS-DMA@PDMS coatings exhibited enhanced biofouling resistance compared with blank surfaces. This superiority in performance could be ascribed to the relatively low surface energy characteristic of the POSS-DMA@PDMS coating, which serves to impede the attachment and proliferation of fouling organisms. Therefore, the coating had a weaker interaction with the organisms. The POSS-DMA@PDMS-TCM coating exhibited an excellent antibiofouling performance, with only a few organisms colonizing it after 120 days. The antibiofouling performance of the coating was attributed to the combined effect of the antifouling agents and the low surface energy. Figure S13 illustrates the antifouling mechanism of the POSS-DMA@PDMS-TCM coating, in which the attached antifoulant (TCM) acted as a contact killer. Because the antifouling agent groups are firmly anchored within the polymer chain, the surface of the coating displays a durable antibiofouling capacity. PDMS segments with low surface energy facilitated biofouling removal. Surface chemistry is a significant factor in the formation and release of fouling organisms. The Behr curve indicates that fouling propensity typically reaches a minimum at critical surface tensions, falling within the range of 22–24 mN/m. Within this specific surface tension interval, the interaction between microorganisms and the surface is significantly reduced.⁶⁹ As shown in Figure S14, the POSS-DMA@PDMS-TCM coating had a critical surface tension of 22.99 mN/m and had the fewest nondestructive organisms attached to its surface. As shown in Table S3, we compared the developed POSS-DMA@PDMS-TCM coating with other recently developed similar materials in the comprehensive marine field tests.^{70–73} Overall, the marine field tests demonstrated that the POSS-DMA@PDMS-TCM coating has promising maritime applications.

3.7. Anticorrosion Properties. When the POSS-DMA@PDMS-TCM coating was applied onto the metallic substrates, it effectively functioned as a protective layer with anticorrosion capabilities. The Fe and Cu samples coated with POSS-DMA@PDMS-TCM were exposed to an acidic aqueous solution (HCl, 3 M), respectively (Figure S15a,b). It was found that the surfaces of the samples coated with POSS-DMA@PDMS-TCM retained their original state. In contrast, the uncoated pure metal surfaces underwent rapid corrosion. As shown in Figure S15c, a pure aluminum boat and an aluminum boat coated with POSS-DMA@PDMS-TCM, both loaded with methylene blue dye, were positioned on the surface of a 3 M HCl solution. The aluminum boat initiated the corrosion process within 3 min and underwent complete corrosion by the 10 min mark. In contrast, no leakage of the blue dye was detected from the aluminum boat coated with POSS-DMA@PDMS-TCM, which indicated that the POSS-DMA@PDMS-TCM coating effectively exerted a protective function on the metal.

Accurately determining metal corrosion rates is important for corrosion protection. In this experiment, all working electrodes' reaction areas could be regarded as the same (with an exposed area of 1 cm²), so the corrosion resistance of each coating could be determined by directly comparing corrosion potential (E_{corr}) and corrosion current density (I_{corr}). The polarization resistance (R_p) values were calculated according to the Stern–Geary equation.⁷⁴

$$R_p = \frac{b_a b_c}{2.303(b_a + b_c)I_{\text{corr}}} \quad (4)$$

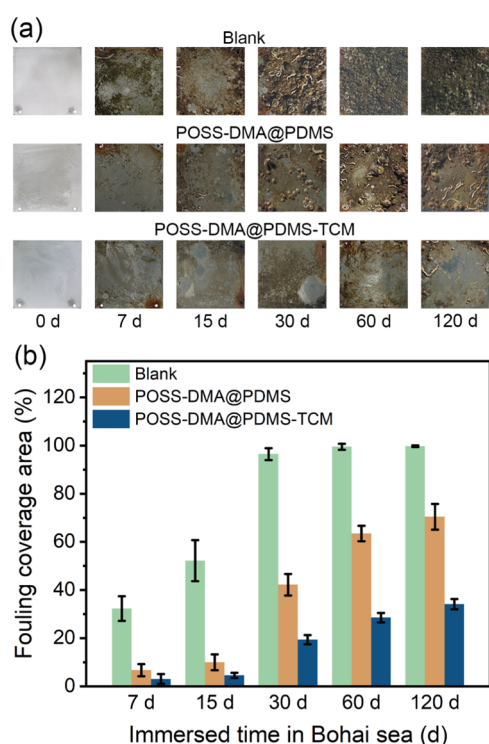


Figure 7. Marine field antifouling test. (a) Images of stainless steel plate samples and (b) quantitative data on the area fouled by fouling organisms after immersion in the Bohai Sea for 7, 15, 30, 60, and 120 days.

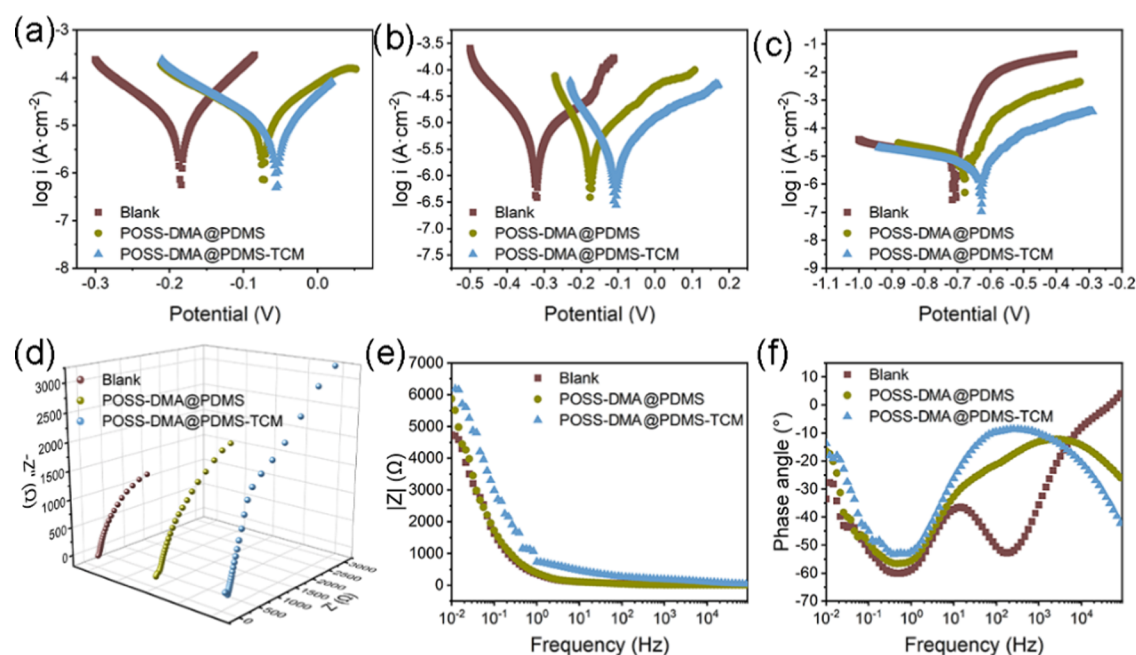


Figure 8. Anticorrosion property analysis. Potentiodynamic curves of different coatings on (a) copper (Cu), (b) iron (Fe), and (c) aluminum (Al) in 3.5 wt % NaCl solution. (d) Nyquist, (e) Bode, and (f) impedance plots of different coatings on Cu in 3.5 wt % NaCl solution for the same time.

where I_{corr} is determined by the intersection of the linear parts of the anode and cathode curves and b_a and b_c are the anodic and cathodic slopes, respectively.

Higher E_{corr} , lower I_{corr} , and higher R_p values generally indicate better corrosion protection.⁷⁸ As shown in Figure 8a–c and Table S4, the control (Cu, Fe, and Al) had the lowest E_{corr} (−0.194, −0.322, and −0.710 V, respectively) and the highest I_{corr} (1.34×10^{-5} , 7.23×10^{-6} , and 3.14×10^{-5} A·cm⁻², respectively), indicating the worst corrosion resistance. The POSS-DMA@PDMS-TCM coating had the highest E_{corr} values of −0.055, −0.109, and −0.624 V, and the lowest I_{corr} of 7.67×10^{-6} , 1.31×10^{-6} , and 1.82×10^{-6} , for Cu, Fe, and Al, respectively. Thus, the POSS-DMA@PDMS-TCM coating had excellent corrosion resistance. The R_p values of the POSS-DMA@PDMS-TCM coating were 3.10×10^5 , 2.37×10^6 , and 1.17×10^6 Ω for Cu, Fe, and Al, respectively, which further confirmed their good barrier performance.

Figure 8d–f shows that the corrosion resistance of the coating was evaluated by EIS measurements. The radius of a high-frequency-tolerant arc is generally considered to be the magnitude of the mass transfer resistance of samples in a Nyquist diagram. Here, a larger mass transfer resistance indicated a slower substrate corrosion rate. As shown in Figure 8e, the POSS-DMA@PDMS-TCM coating had the largest radius of the high-frequency capacitive resistance arc and, thus, the slowest corrosion rate. This was further confirmed by the impedance and Porter-phase plot. The impedance plot showed that the longitudinal axis intercept of the POSS-DMA@PDMS-TCM coating curve was the largest. According to previous studies, the intercept between the curve and the $\log|Z|$ axis decreases with the degradation of a coating on a metal surface. Furthermore, the plot showed a decreasing longitudinal axis intercept of the modulus curve. Therefore, at the same time and under the same environment, the degradation rate of the POSS-DMA@PDMS-TCM coating was the smallest. The Bode phase plot, or the electrochemical frequency–phase ($\log$ -phase angle) curve, relied on three

key parameters to explain the relationship between its characteristics and the actual corrosion performance: the minimum phase angle φ_{min} , the breakpoint frequency f_b , and the corresponding frequency f_{min} at the minimum frequency. The POSS-DMA@PDMS-TCM coating had the lowest degradation rate at the same time and in the same environment. It also had the highest φ_{min} and lower f_b and f_{min} , indicating that this coating had excellent corrosion resistance. The experimental data was fitted using an equivalent circuit model (Figure S15). In this circuit, R_s stands for the solution resistance, Q_c is the coating capacitance, and R_c is the coating resistance. In other words, when coated on metallic substrates, the POSS-DMA@PDMS-TCM coating acts as a barrier that protects the metal from corrosion.

In addition, we analyzed the long-term corrosion resistance of the coatings by means of Tafel curves. Figure S17 illustrates the Tafel polarization curves of the metals, POSS-DMA@PDMS coating, and POSS-DMA@PDMS-TCM immersion in artificial seawater after 7 and 15 days. Tables S5 and S6 show the equivalent data for polarization restrictions obtained through Tafel extrapolation. As shown in Figure S17a–c and Table S5, after 7 days of immersion in artificial seawater, the corrosion currents of pure metal ($I_{\text{corr}} = 5.77 \times 10^{-6}$, 4.61×10^{-6} , and 3.35×10^{-5} for Cu, Fe, and Al, respectively) and POSS-DMA@PDMS coatings ($I_{\text{corr}} = 6.45 \times 10^{-6}$, 1.69×10^{-6} , 1.28×10^{-5} for Cu, Fe, and Al, respectively) were higher than those of POSS-DMA@PDMS-TCM coatings ($I_{\text{corr}} = 2.31 \times 10^{-6}$, 1.13×10^{-6} , 7.34×10^{-6} for Cu, Fe, and Al, respectively). As shown in Figure S17d–f and Table S6, after 15 days of immersion in artificial seawater, the corrosion currents of pure metal ($I_{\text{corr}} = 6.78 \times 10^{-6}$, 3.21×10^{-6} , and 1.04×10^{-4} for Cu, Fe, and Al, respectively) and POSS-DMA@PDMS coatings ($I_{\text{corr}} = 4.77 \times 10^{-6}$, 2.27×10^{-6} , 9.68×10^{-6} for Cu, Fe, and Al, respectively) were higher than those of POSS-DMA@PDMS-TCM coatings ($I_{\text{corr}} = 2.04 \times 10^{-6}$, 1.62×10^{-6} , 9.24×10^{-6} for Cu, Fe, and Al, respectively). In addition, the corrosion current (E_{corr}) of the coating was

highest after immersion in artificial seawater for 7 or 15 days. These results indicated that POSS-DMA@PDMS-TCM coatings exhibit superior long-term corrosion resistance.

The enhanced corrosion protection of the metals coated with POSS-DMA@PDMS-TCM can be attributed to the combined influence of the hydrophobic nature and the specific chemical properties inherent to the coating. As shown in Figure S18, the hydrophobicity of the coating creates a barrier between the metal substrate and the corrosive medium, thereby slowing or inhibiting the migration of corrosive particles. In addition, DMA is a corrosion inhibitor that retards further corrosion processes by forming a passive layer on metal surfaces through catechol-Me coordination bonds.⁷⁶

4. CONCLUSIONS

In summary, a novel POSS-DMA@PDMS-TCM coating was successfully synthesized using a thiol–ene click reaction. DMA and POSS were demonstrated to improve the adhesion and mechanical strength of the PDMS coatings. The strong adhesion (2.17 MPa in air and 2.11 MPa underwater) of the resulting coating was achieved. It also exhibited strong mechanical stability under tape peeling, sandpaper abrasion cycles, and water jetting. The POSS-DMA@PDMS-TCM coating achieved antifouling rate of approximately 98.1% for *S. aureus*, 99.5% for *E. coli*, and 94.5% for diatoms, indicating its good performance in preventing the adhesion of marine microorganisms. The effectiveness of the POSS-DMA@PDMS-TCM coating in preventing the attachment of marine fouling organisms was further demonstrated by 120 days of antifouling testing in real marine environments. In addition, Tafel polarization and EIS measurements demonstrated that the POSS-DMA@PDMS-TCM coating had outstanding anticorrosion properties for the metal substrates. This work provides a feasible and long-term solution for marine antifouling and anticorrosion.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, P.V., upon reasonable request.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c18569>.

Physical drawings of coating materials on different substrates; morphology characterization of the DMA@PDMS coating and POSS-DMA@PDMS; application of the POSS-DMA@PDMS-TCM coating as adhesive materials; roughness of the POSS-DMA@PDMS coatings with different DMA contents; the POSS-DMA@PDMS-TCM coating displayed super-repellency to aqueous solutions with different pH values (pH = 1, 3, 5, 7, 9, 11, and 13); the variation of water CA values after tape peeling, sandpaper abrasion cycles, sand abrasion cycles, and water jetting on two different coating surfaces; images of panels (epoxy resin) of marine field assessment, and statistics of fouling coverage area after immersion in Bohai Sea for 7, 15, 30, 60, and 120 days; photos of panels (stainless steels) processed by ImageJ software; photos of panels (epoxy resin) processed by ImageJ software; schematic diagram of the antifouling procedures for POSS-DMA@PDMS-TCM coatings; SFE calculated by average WCA and

DCA; fitted equivalent circuit model of the copper and the POSS-DMA@PDMS-TCM coating; the anticorrosion performance of the Fe, Cu, and Al with and without POSS-DMA@PDMS-TCM coating under 3 M HCl solution; schematic diagram of the corrosion procedures for metal coated with POSS-DMA@PDMS-TCM coating; a comparison of the pull-off adhesion of our work to some reported works; a comparison of the shear strength of our work to some reported works; a summary of reported marine field test results of antifouling coatings; and values of E_{corr} , I_{corr} , and R_p for the copper, iron, and aluminum samples coating with and without the PDMS-based coating in 3.5 wt % NaCl solution (PDF)

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Notes

The authors declare no competing financial interest.

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